1. Auxins promote the **elongation of stems** and **coleoptiles** which makes them to grow.

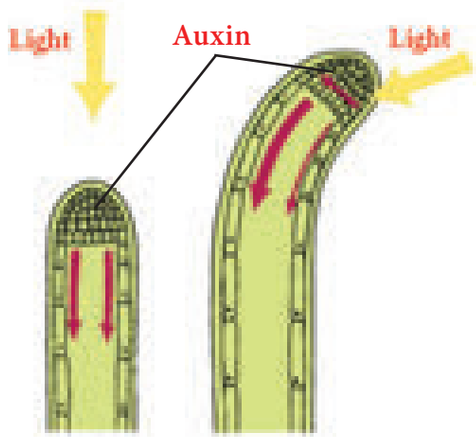


Figure 16.3 Cell Elongation

2. Auxins **induce root formation** at low concentration and inhibit it at higher concentration.
3. The auxins produced by the apical buds suppress growth of lateral buds. This is called **apical dominance**.
4. Seedless fruits without fertilization are induced by the external application of auxins. (**Parthenocarpy**). Examples: Watermelon, Grapes, Lime etc.
5. Auxins **prevent** the formation of **abscission layer**.



Phenyl Acetic Acid (PAA) and Indole 3 Acetonitrile (IAN) are natural auxins. Indole 3 Butyric Acid (IBA), Indole-3-Propionic Acid, α-Naphthalene Acetic Acid (NAA), 2, 4, 5-T (2,4,5 Trichlorophenoxy Acetic Acid) are some of the synthetic auxins.

16.1.2 Cytokinins

Cytokinins (Cytos - cell; kinesis - division) are the plant hormones that **promote cell division** or cytokinesis in plant cells. It was first isolated from Herring fish sperm. Zeatin was the cytokinin isolated from *Zea mays*. Cytokinin is found abundantly in liquid endosperm of coconut.

Physiological effects of cytokinins

1. Cytokinin induces **cell division** (cytokinesis) in the presence of auxins.
2. Cytokinin also causes **cell enlargement**.
3. Both auxins and cytokinins are essential for the formation of new organs from the callus in tissue culture (**Morphogenesis**).
4. Cytokinins promote the growth of **lateral buds** even in the presence of apical bud.
5. Application of cytokinin delays the process of ageing in plants. This is called **Richmond Lang effect**.

16.1.3 Gibberellins

Gibberellins are the most abundantly found plant hormones. Kurosawa (1926) observed **Bakanae disease** or **foolish seedling disease** in rice crops. This internodal elongation in rice was caused by fungus *Gibberella fujikuroi*. The active substance was identified as **Gibberellic acid**.

Physiological effects of gibberellins

1. Application of gibberellins on plants stimulate extraordinary **elongation of internode**. e.g. Corn and Pea.

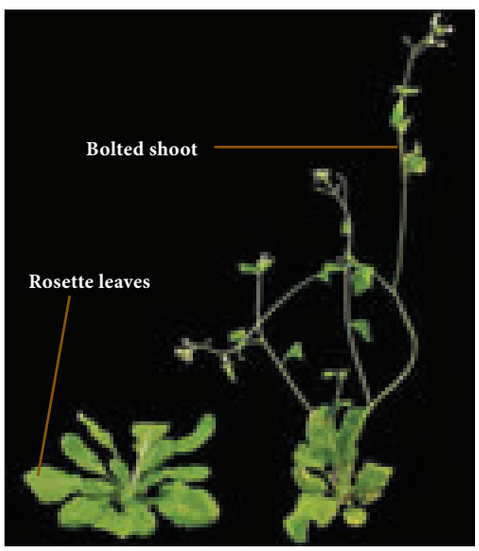


Figure 16.4 Bolting

2. Treatment of rosette plants with gibberellin induces sudden shoot elongation followed by flowering. This is called **bolting**.

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